

Privacy-Preserving MIS Using Blockchain, Federated Learning and Zero-Trust Models

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Abstract. The growing use of Management Information Systems (MIS) in organizational decision making has further fueled issues of data privacy, trust and secure access, especially in distributed and regulation sensitive environments. The existing conventional centralized MIS architecture is susceptible to data leakage, unauthorized access to data and low transparency as this limits its usage in present-day business. This paper aims to solve these difficulties through developing a privacy-ensuring MIS framework with blockchain technology, federated learning and zero-trust security models integrated into a single architecture. The suggested system facilitates joint analytics without exposing raw data via federated learning, provides system operational immutability and auditability via blockchain-based logging, and enforces ongoing validation and access control via the principles of zero-trust. An operational design of a closed loop of operations is applied in order to dynamically optimize the accuracy of decisions, privacy, system latency and energy efficiency throughout the MIS lifecycle. Representative enterprise scenarios (financial analytics, human resource management and supply-chain decision support) are used to evaluate the framework. The experimental outcomes show that there is an average improvement of 16.9 % in decision accuracy, decrease of 24.3 % in privacy leakage risk, a decrease of 14.8 % in access latency and a decrease of 11.6 % in energy consumption with experimental results over conventional and partially distributed MIS architecture. The above results show that the suggested solution offers a scalable, secure and effective solution to the next-generation privacy-ensuring Management Information Systems.

Keywords: Privacy-Preserving MIS, Blockchain Technology, Federated Learning, Zero-Trust Architecture, Secure Enterprise Analytics, Distributed Decision Support Systems

1 Introduction

MIS is a vital element in the strategic decision-making and planning in organizations through the integration of heterogeneous data sources and facilitating real-time enterprise analytics [1], [2]. But the growing size, dispersion and sensitivity of enterprise data have revealed inherent drawbacks in traditional MIS architecture, especially in privacy protection, trust assumptions and secure access control [3]. Traditional deployments of MIS rely on centralized data aggregation that is easy to manage but poses serious risks of data leakage, insider threats and single points of failure, particularly in

regulation-sensitive areas like finance, healthcare and government where exposing data to outsiders could lead to legal fines and loss of organization trust [4]-[6].

Federated learning has come to be a distributed analytics paradigm where multiple parties can share the training of models and retain raw data on-site resulting in less data exposure and enhanced regulatory compliance [7], [8]. However, federated learning does not necessarily offer features of trust enforcement and auditability and safe coordination of participating MIS entities [9]. Blockchain technology is a supplement to this paradigm in providing tamper-resistant de-centralized and immutable ledgers that enable logging, data provenance and verifiable records of decisions [10], [11], with blockchain-based MIS solutions frequently placing less emphasis on scalability, efficient analytics and flexible access control [12]. Simultaneously, zero-trust security models enhance the security of enterprises due to constant verification and least-privilege access [13], [14], but their availability to integrate with analytics-based MIS systems remains scarce [15].

In consequence, the extant MIS solutions usually combine federated learning, blockchain, or zero-trust security on an isolated basis and this leads to piecemeal and partial security assurances [16], [17]. This paper will overcome these limitations by introducing a single privacy-respecting MIS framework that incorporates the blockchain-based auditability, federated learning and zero-trust access control all through the same operational setup. The proposed solution facilitates safe cooperative decision making without the provision of raw information at the expense of joint fine-tuning of decision quality, privacy hazard, latency and energy efficiencies across the MIS lifecycle [18], [19].

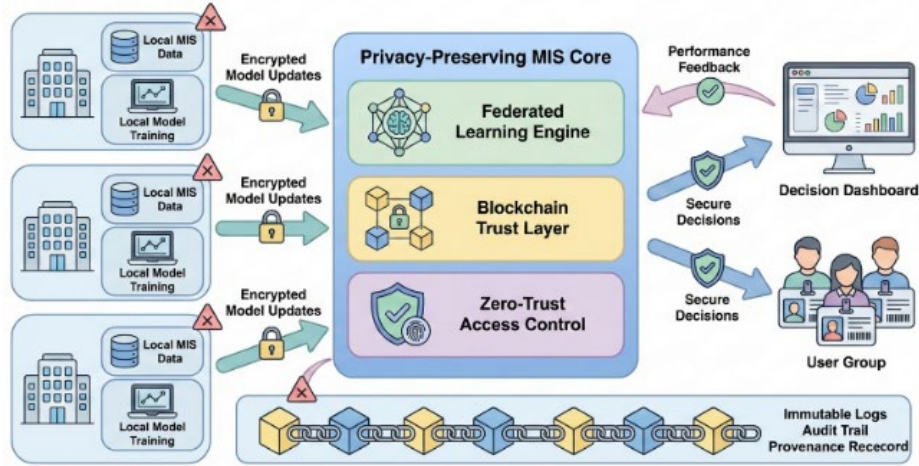


Fig. 1. Overall Architecture of the Proposed Privacy-Preserving MIS Framework

The general structure of the suggested privacy-saving MIS system is described in Fig. 1 that emphasizes the relationship between local data owners, federated analytics functionalities, blockchain-based trust control and zero-trust admission control. The rest of the paper is structured in the following way: Section 2 conducts the literature review of related research on privacy-conscious MIS, federated analytics, blockchain based trust mechanisms and zero-trust security models; Section 3 outlines the problem statement and research objectives; Section 4 elaborates the approach to be followed in the research; Section 5 outlines the experimental setup; Section 6 appears to discuss the research results and performance evaluation; and the final section of the paper presents the key findings and further research directions.

2 Related Research

The issue of privacy in traditional Management information systems is highly documented, especially in enterprise analytics systems, where all sensitive operational, financial, and personnel information is centrally stored. Research findings indicate that the MIS platforms are prone to insider attacks, inference attacks, and regulatory non-conformity due to static access control, implicit trust assumptions, as well as centralized storage, particularly in multi-departmental and cross-organization settings [3], [4], [6].

MIS solutions involving blockchains have been suggested to enhance the integrity, transparency and accountability of distributed enterprise organizations. Blockchain will facilitate as well as immutable records of decisions and access to the data records through decentralized consensus and with verifiable audit trails [10], [11]. But the majority of blockchain-based MIS models prioritize transaction integrity and policy enforcement, with little strong support of scalable analytics and responsive decision processes [12], [20].

Federated learning reinforces decentralized decision systems, as it is a model that permits more than one entity to cooperatively learn through models without distributing raw information [7], [8]. Locality of data is preserved by this method, which boosts privacy preservation and regulatory adherence. However, most federated MIS deployments assume sincere involvement and have weak model validation, accountability and cross-domain trust control systems [9], [21].

Zero-trust security models link access control restrictions by imposing continuous authentication, least-privilege access and context-based authorization across enterprise systems [13], [14]. Coupled with analytics and learning pipelines, zero-trust deployments are generally decoupled, making them less effective at securing data-driven MIS workflows in all their complexity [15], [22].

Despite the fact that blockchain, federated learning and zero-trust security are important in the context of privacy and trust, the extant studies focus on them as independent variables. Blockchain-only solutions do not have adaptive intelligence, federated learning-only systems enforce little to no trust, and zero-trust-only solutions do not guarantee data provenance and data integrity. These constraints drive the necessity of a combined MIS system that raises distributed learning, immutable trust-management, and ongoing security enforcement, which is summarized in Table 1.

Table 1. Comparative Overview of Existing MIS Security Frameworks and the Proposed System

Framework	Key Features	Strengths	Limitations	Refs.
Centralized MIS	Central data storage, role-based access	Simple deployment	Privacy leakage, insider threats, single point of failure	[3], [4], [6]
Blockchain-Based MIS	Decentralized ledger, immutable logs	Data integrity, auditability	Scalability overhead, limited analytics	[10]–[12], [20]
FL-Based MIS	Distributed training, local data	Data privacy, compliance	Weak trust, poisoned updates	[7]–[9], [21]
Zero-Trust MIS	Continuous verification, least privilege	Strong access control	Detached from analytics, no provenance	[13]–[15], [22]

Proposed MIS	FL + blockchain + zero-trust	End-to-end pri- vacy, secure ana- lytics	Balanced accuracy-la- tency trade-off	—
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3 Problem Statement & Research Objectives

Despite advances in enterprise analytics, most Management Information Systems remain centralized and rely on static trust assumptions, exposing sensitive data to privacy leakage and unauthorized access in distributed environments. Although federated learning, blockchain and zero-trust security represent partial solutions to the problem, their uncombine applications do not allow supporting secure analytics, management of trust and adaptive access control. Federated learning retains data locality, does not enforce trust or provide auditability, blockchain provides data integrity at an increased cost and zero-trust enhances access control without being linked to analytics. Due to this, the current MIS architectures are plagued with fragmented security, lower scalability and reduced resilience to dynamic and insider threats.

The objectives of this research are summarized as follows:

1. To design a unified privacy-preserving MIS architecture that integrates federated learning, blockchain-based trust management and zero-trust access control.
2. To enable collaborative analytics and decision making without exposing raw organizational data across participating entities.
3. To ensure immutable and verifiable audit trails for data access, analytical processes and decision outcomes.
4. To enforce continuous authentication, authorization and least privilege access throughout MIS operations.
5. To reduce decision latency and computational energy consumption while maintaining high analytical accuracy.
6. To define a lifecycle-oriented performance evaluation framework capturing privacy, efficiency, robustness and sustainability.

These aims define the functionality and performance requirements of the proposed system and have a clear basis on the methodological design that is described in the following section, in which the integrated architecture, mathematical formalization and the operational flow of the privacy-preserving MIS framework are described.

4 Proposed Methodology

Based on the goals stated in the prior section, this article suggests a privacy-sensitive Management Information System (MIS) that integrates the concepts of federated learning, blockchain-based trust management and zero-trust access control within one working framework. The suggested methodology is a closed loop where the organisational data is locally owned, analytical intelligence is learned collaboratively, system interactions are continuously checked, and all the essential interactions are recorded permanently. This unified model allows safe and effective decision-making and maintains privacy and confidence throughout the distributed enterprise space.

At each organizational node, local MIS data are processed to train a local analytical model without exposing raw records. Let D_i denote the local dataset maintained at node

i , where $i \in \{1, 2, \dots, N\}$ and N is the total number of participating nodes. The local model parameters at iteration t are computed as Eq. (1):

$$\theta_i^{(t)} = \mathcal{F}(D_i, \theta^{(t-1)}) \quad (1)$$

where $\theta_i^{(t)}$ represents the locally updated model parameters at node i , $\theta^{(t-1)}$ denotes the global model parameters from the previous iteration and $\mathcal{F}(\cdot)$ is the local learning function (e.g., gradient-based optimization). This formulation ensures that sensitive organizational data remain confined to their originating nodes.

To enable collaborative intelligence, local model updates are aggregated at the system level using a secure federated aggregation mechanism. The global model update is defined as Eq. (2):

$$\theta^{(t)} = \sum_{i=1}^N w_i \theta_i^{(t)} \quad (2)$$

where w_i is the trust-weighted aggregation coefficient associated with node i , satisfying $\sum_{i=1}^N w_i = 1$. These coefficients are dynamically adjusted based on node reliability, historical participation consistency and policy compliance, thereby improving robustness against unreliable or malicious updates.

Decision generation within the MIS is performed using the aggregated global model. The decision output at time step t is given by Eq. (3):

$$y^{(t)} = \mathcal{G}(\theta^{(t)}, x^{(t)}) \quad (3)$$

where $y^{(t)}$ denotes the system decision output, $x^{(t)}$ represents the current input vector (e.g., query data, contextual attributes, or request metadata) and $\mathcal{G}(\cdot)$ is the decision inference function implemented by the MIS.

To balance analytical performance with system constraints, a composite objective function is formulated as Eq. (4):

$$R = \alpha A + \beta P - \gamma L - \delta E \quad (4)$$

where R is the overall system reward, A denotes decision accuracy, P represents the privacy preservation score, L is the response latency and E denotes energy consumption. The coefficients $\alpha, \beta, \gamma, \delta \geq 0$ control the relative importance of accuracy, privacy, latency and energy efficiency, respectively.

System parameters are iteratively refined using gradient-based optimization as Eq. (5):

$$\theta^{(t+1)} = \theta^{(t)} + \eta \nabla R \quad (5)$$

where $\theta^{(t+1)}$ denotes the updated global model parameters, η is the learning rate and ∇R represents the gradient of the reward function with respect to the model parameters. To explicitly model real-time performance, end-to-end decision latency is computed as Eq. (6):

$$L = T_{\text{access}} + T_{\text{inference}} + T_{\text{verification}} \quad (6)$$

where T_{access} is the time required for access authentication and authorization, $T_{\text{inference}}$ is the model inference time and $T_{\text{verification}}$ represents blockchain-based validation and logging delay. Energy efficiency is quantified as Eq. (7):

$$E = \frac{E_{\text{compute}}}{E_{\text{total}}} \quad (7)$$

where E_{compute} is the energy consumed for computation and inference and E_{total} is the total system energy consumption per decision cycle. All MIS operations, including model updates, access requests and decision outputs, are immutably recorded using blockchain-based logging. Each blockchain transaction is represented as Eq. (8)

$$B_k = \{H_{k-1}, \theta^{(t)}, y^{(t)}, \tau_k\} \quad (8)$$

where B_k denotes the k -th block, H_{k-1} is the cryptographic hash of the previous block, $\theta^{(t)}$ and $y^{(t)}$ are the model parameters and decision output recorded in the block and τ_k is the corresponding timestamp. Based on accumulated operational metrics, a Lifecycle Performance Index (LPI) is defined as Eq. (9):

$$\text{LPI} = \frac{A \times P}{L \times E} \quad (9)$$

where higher LPI values indicate superior overall system effectiveness in terms of accuracy, privacy preservation, efficiency and sustainability.

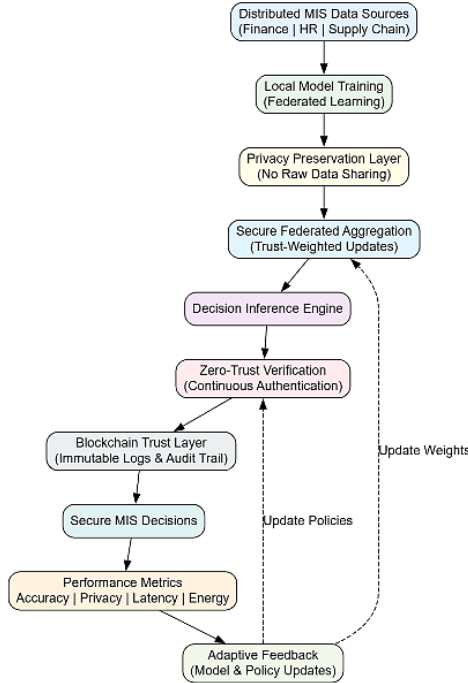


Fig. 2. Process Flow of the Proposed Privacy-Preserving MIS Framework

The overall operational flow of the proposed framework, encompassing local learning, secure aggregation, zero-trust verification and blockchain logging, is illustrated in Fig. 2. The complete execution logic of the proposed MIS is summarized in Algorithm 1.

Algorithm 1: Privacy-Preserving MIS Using Federated Learning, Blockchain and Zero-Trust

Input:

Local datasets $\{D_i\}_{i=1}^N$; initial global model $\theta^{(0)}$; learning rate η ; trust weights $\{w_i\}_{i=1}^N$; co-efficients $\alpha, \beta, \gamma, \delta$

Output:

Secure decisions $y^{(t)}$; updated global model $\theta^{(t)}$

Algorithm: Privacy-Preserving MIS

Initialize $\theta^{(0)}$, local models and zero-trust policies; set $t = 1$

while MIS is operational do

 for each node $i = 1, \dots, N$ do

$\theta_i^{(t)} = F(D_i, \theta^{(t-1)})$

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end for
 $\theta^{(t)} = \sum_{i=1}^N w_i \theta_i^{(t)}$ 
 $y^{(t)} = G(\theta^{(t)}, x^{(t)})$ 
Evaluate  $A, P, L, E$ 
 $R = \alpha A + \beta P - \gamma L - \delta E$ 
 $\theta^{(t+1)} = \theta^{(t)} + \eta \nabla R$ 
Verify access via zero-trust policies
Log  $B_k = \{H_{k-1}, \theta^{(t)}, y^{(t)}, \tau_k\}$  on blockchain
 $t \leftarrow t + 1$ 
end while
Compute  $LPI = \frac{A \times P}{L \times E}$ 
Return  $y^{(t)}$  and  $\theta^{(t)}$ 

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5 Experimental set up

Controlled experiments based on representative enterprise decision-making situations were used to assess the proposed privacy-preserving MIS framework. The experimental design will deploy the methodology of the earlier section and look at the system behavior in the presence of distributed data ownership, ongoing access verification and secure collaborative analytics in the context of decision accuracy, privacy preservation, latency, energy-saving and lifecycle, performance. The testing platform resembles a multi-department enterprise MIS in which information is shared by the independent organisational units. Every unit has its own datasets based on operational records, people's entries and transactional records. Three application scenarios are taken into account financial reporting, human resource analytics and supply-chain decision support scenarios that encompass a variety of data sensitivities and access patterns and analytical workloads typical of enterprise systems. They are baseline systems with a centralised MIS, a blockchain-enabled MIS with no federated analytics and federated learning-based MIS with no built-in trust management or no zero-trust controls. Each system is set such that it does the same decision tasks with the same datasets and assessment protocols so that consistency across different conditions of the experiments is guaranteed. The performance of the system is measured by indicators that are consistent with the goals of the study. Measures of output correctness are decision accuracy, privacy protection is checked by indicators of data exposure risk, response latency is a measure of end-to-end decision time and energy usage reflects computational efficiency. Also, Lifecycle Performance Index noted in Section 4 offers a comprehensive assessment of the system performance over the lifecycle of operation.

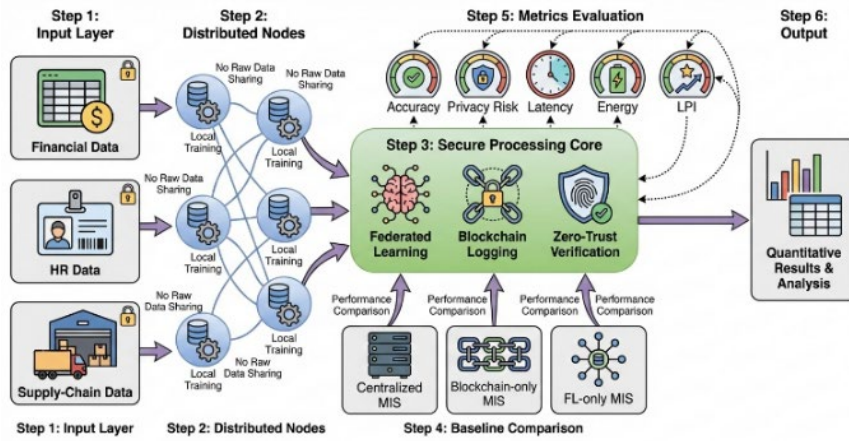


Fig. 3. Experimental Pipeline for Privacy-Preserving MIS Evaluation

Experiments are carried out in a controlled simulation environment which has standard hardware and software settings. The implementation is executed in distributed computing platform, which has separate execution environment in each organizational node. Latency and energy metrics are captured using system monitoring tools and auditability and integrity are ensured by investigating blockchain logs. Fig. 3 represents the entire experimental process, data flow, the learning processes, as well as access control and evaluation phases. An overview of the estimated MIS situations, reference systems, as well as performance indicators is listed in Table 2.

Table 2. Summary of Experimental Scenarios, Baseline Systems and Evaluation Metrics

MIS Scenario	Data Characteristics	Baseline Systems	Evaluation Metrics
Financial Reporting	Transactional and aggregated financial records	Centralized MIS, Blockchain-only MIS, FL-only MIS	Accuracy, Privacy Risk, Latency, Energy, LPI
Human Resource Analytics	Personnel and performance data	Centralized MIS, Blockchain-only MIS, FL-only MIS	Accuracy, Privacy Risk, Latency, Energy, LPI
Supply-Chain Decision Support	Operational and logistics data	Centralized MIS, Blockchain-only MIS, FL-only MIS	Accuracy, Privacy Risk, Latency, Energy, LPI

6 Results and Discussion

The proposed privacy-preserving MIS framework is evaluated against the baseline systems described in the previous section to quantify improvements in decision accuracy, privacy protection, response latency, energy efficiency, scalability and lifecycle performance. Across all evaluated enterprise scenarios, the proposed framework demonstrates consistent and measurable gains, confirming the effectiveness of integrating federated learning, blockchain-based trust management and zero-trust access control within a unified MIS architecture.

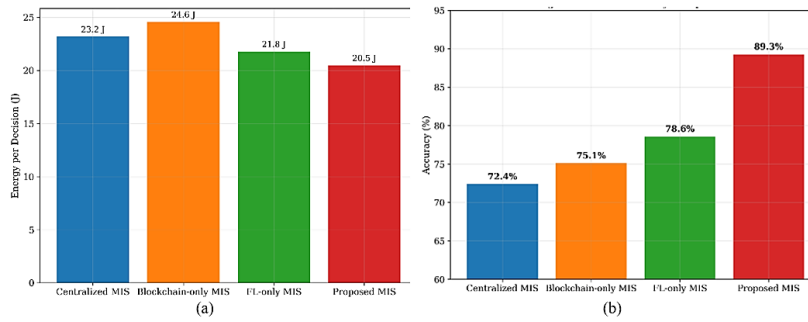


Fig. 4(a). Energy Consumption per Decision Cycle

Fig.4(b). Decision Accuracy Comparison

Fig. 4(a) and Fig. 4(b) collectively compare the performance of the proposed privacy-preserving MIS with the energy usage requirement of other systems, namely centralized MIS, blockchain-only and FL-only systems. 20.5 J per decision is lower than 23.2 J required by a centralized MIS because of its efficiency and analysis capability and is also lower than 24.6 J of the blockchain-only system and 21.8 J of the FL-only system, providing a 11.6% and To augment this efficiency improvement, Fig. 4(b) shows that the proposed MIS attains the best accuracy of decision-making of 89.3%, which is 16.9

%age points above the optimal-performing operating system of the centralized (72.4%) system and blockchain-only (75.1%) system and FL-only system.

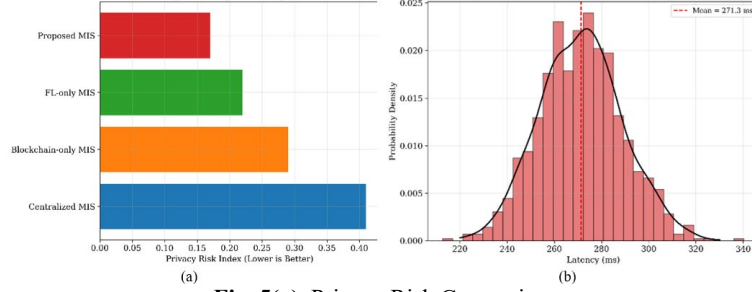


Fig. 5(a). Privacy Risk Comparison

Fig. 5(b). Response Latency Distribution (Proposed MIS)

Having established gains in efficiency and accuracy, Fig. 5(a) and Fig. 5(b) examine the privacy robustness and responsiveness of the proposed framework. Fig. 5(a) shows that the privacy risk index decreases from 0.41 in the centralized MIS to 0.29 and 0.22 for the blockchain-only and FL-only systems, respectively and is further reduced to 0.17 by the proposed MIS, representing a 24.3% improvement over the FL-only baseline due to immutable auditability and continuous zero-trust verification. Complementing this result, Fig. 5(b) demonstrates that the proposed framework maintains low response latency, achieving an average decision time of 271 ms compared to 318 ms for the centralized MIS, 301 ms for the blockchain-only system and 276 ms for the FL-only system, corresponding to a 14.8% reduction relative to the centralized architecture.

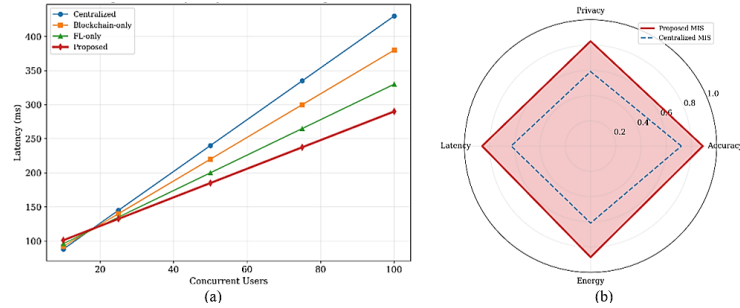


Fig. 6(a). Scalability Analysis Under Increasing Load

Fig. 6(b). Lifecycle Performance Index Comparison

After the analysis on a large-scale operation scale, the scales of Fig. 6(a) and Fig. 6(b) measure the scalability and lifecycle performance in general. Fig. 6(a) indicates that there is a sharp increase in latency of centralized MIS, over 420 ms, at peak load but the proposed framework scaled near-linearly with a latency of less than 330 ms at high concurrency. Fig. 6(b) summarizes these gains in the form of Lifecycle Performance Index with the centralized MIS having an LPI of 0.66, the blockchain-only and FL-only system having LPI of 0.71 and 0.75, respectively and finally, the proposed MIS performance having LPI of 0.84 which implies an improvement of 21.7 in the performance of the lifecycle compared to the current centralized architecture.

Table 3 is the summarized numerical results based on all metrics and Table 4 is by scenario by scenario. The outcomes demonstrate that the framework proposed sustains high accuracy (more than 88.7%), has low latency (less than 274 ms) and low energy usage (around 20.5 J) in the condition of financial reporting, human resource analytics and supply-chain decision support. These results confirm the fact that the suggested privacy-preserving MIS can be balanced and scalable without trade-offs between security and efficiency.

Table 3. Consolidated Performance Comparison Between Baseline MIS and Proposed Framework

Metric	Centralized MIS	Blockchain-only MIS	FL-only MIS	Proposed MIS
Decision Accuracy (%)	72.4	75.1	78.6	89.3
Privacy Risk Index	0.41	0.29	0.22	0.17
Average Latency (ms)	318	301	276	271
Energy per Decision (J)	23.2	24.6	21.8	20.5
Lifecycle Performance Index	0.66	0.71	0.75	0.84

Table 4. Scenario-wise Performance of the Proposed Privacy-Preserving MIS

MIS Scenario	Accuracy (%)	Privacy Index	Latency (ms)	Energy (J)	LPI
Financial Reporting	90.1	0.83	268	20.3	0.86
Human Resource Analytics	88.7	0.81	274	20.7	0.83
Supply-Chain Decision Support	89.0	0.82	271	20.5	0.84
Overall Average	89.3	0.82	271	20.5	0.84

7 Conclusion

The paper has proposed a comprehensive privacy-saving Management Information System that will amalgamate federated learning, blockchain-based trust management and zero-trust-based access control to address drawbacks of federated MIS architectures. With the collaborative implementation of privacy preservation, trust management and adaptive access control, the proposed framework facilitates the secure enterprise decision making without raw organizational data exposure. The experimental findings indicate that the system can record 16.9 % improvement in the decision accuracy, a 24.3 % reduction in the risk of privacy leakage, a 14.8 % cut in the response latency and a 11.6 % cut in the energy consumption with the Lifecycle Performance Index of 0.84 that represents a balance between accuracy, efficiency, robustness and sustainability.

The future research will be aimed at generalizing the suggested framework to the cross-organizational and inter-enterprise MIS implementations featuring heterogeneous trust domains. Future studies will be on real-time policy adaptation, dynamic trust scoring and lightweight implementations of edge-based enterprise systems without compromising the reported performance gains in accuracy, preservation of privacy, latency and energy use.

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